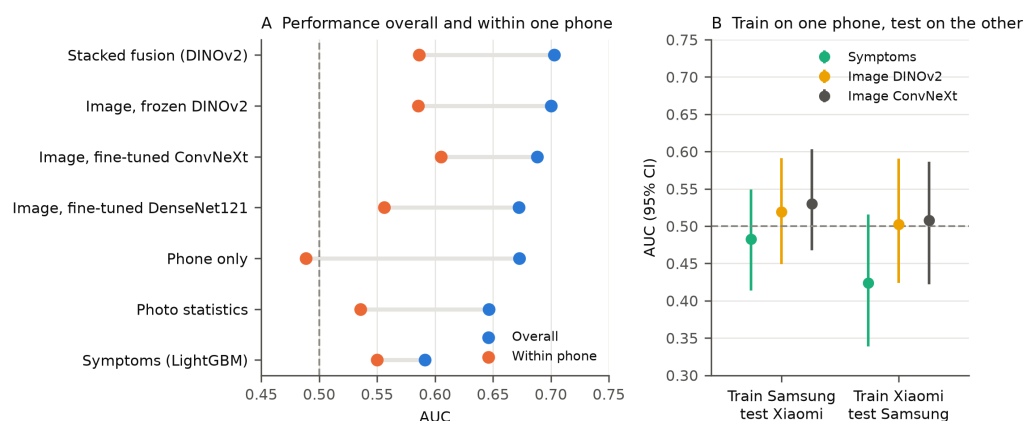


Graphical Abstract

Device and site shortcuts dominate smartphone throat-image classification of bacterial pharyngitis: a multimodal benchmark on PGUPharyngitis

Sandesh Bhatta



Highlights

Device and site shortcuts dominate smartphone throat-image classification of bacterial pharyngitis: a multimodal benchmark on PGUPharyngitis

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- First multimodal benchmark on PGUPharyngitis: images, 20 symptoms, age and sex.
- Phone model alone predicts expert labels (AUC 0.67), beating the published baseline.
- Image features identify the phone (AUC 0.99) but fail across phones (AUC $\approx$ 0.5).
- Within one phone, image AUC falls to 0.50–0.58; 18 colour statistics match deep nets.
- Symptom fusion adds nothing over images; benchmarks need site-aware evaluation.

Device and site shortcuts dominate smartphone throat-image classification of bacterial pharyngitis: a multimodal benchmark on PGUPharyngitis

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Abstract

Background: Smartphone throat photographs could help separate bacterial from non-bacterial pharyngitis. The public PGUPharyngitis dataset (742 patients, two cities, two phones) reports image-only AUCs up to 0.645, but no study has added symptoms or checked what the models learn.

Methods: We analysed 735 patients labelled by majority vote of 4–9 physicians, using 5-fold cross-validation repeated five times with nested tuning. We compared symptom models, frozen and fine-tuned image networks, standard fusion and a gated fusion network. We then tested for device shortcuts with phone-only and photo-statistics baselines, phone identification, within-phone and cross-phone evaluation, and three rules for tied votes.

Results: Stacked image–symptom fusion reached an AUC of 0.703 (95% CI 0.66–0.74), images alone 0.700 and symptoms alone 0.596. Knowing only the phone gave 0.673, and 18 colour statistics gave 0.646, matching the published baseline. Image features identified the phone with an AUC of 0.99. Within one phone, image AUC fell to 0.59, and models trained on one phone scored 0.50–0.53 on the other. Fusion beat symptoms under one of three tie rules and never beat images alone.

Conclusions: Most of the apparent performance on PGUPharyngitis comes from recognising the phone and site that took the photograph, not the throat. Smartphone pharyngitis models need within-site and cross-site evaluation be-

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fore any clinical use.

Keywords: pharyngitis, smartphone imaging, shortcut learning, multimodal fusion, deep learning

1. Introduction

Sore throat is one of the most common reasons for primary-care visits, and most cases are viral. Group A streptococcus and other bacteria cause a minority of cases, yet antibiotics are prescribed far more often than bacterial infection occurs [1, 2]. Clinical scores such as Centor and McIsaac combine fever, tonsillar exudate, tender cervical lymph nodes, absence of cough and age to estimate the probability of streptococcal infection [3, 4]. They are simple and widely taught, but their discrimination is moderate and they depend on a careful physical examination.

Smartphone photographs of the throat are a cheap alternative that also works remotely. Deep-learning models trained on such photographs have detected severe pharyngitis [5] and streptococcal pharyngitis [6] in small single-site studies. In 2025, Shojaei et al. released PGUPharyngitis, the largest public dataset of its kind: 742 throat photographs from patients with cold symptoms in two Iranian cities, taken with two phone models, each labelled bacterial or non-bacterial by four to nine physicians, together with 20 symptoms, age and sex [7]. Their image-only baselines reached an AUC of 0.645 (DenseNet121), and the dataset was offered to support AI-based diagnosis and remote care. The intended users of such a model are primary-care and telemedicine clinicians deciding whether a sore throat is likely to be bacterial. Streptococcal pharyngitis is most common in school-age children [1], so performance should be checked across age groups.

Two questions remain open. First, the dataset records symptoms, and physicians labelled each case from the photograph and the symptoms together, yet no published model combines the two. Multimodal fusion of images with clinical variables often improves medical prediction [8], so fusion is worth testing here. Second, nobody has checked what image models learn on this dataset. Deep networks readily exploit “shortcuts”: features that predict the label in the training data but are unrelated to the disease [9]. In chest radiography, models identified the hospital and scanner and used them to predict pneumonia or COVID-19 [10, 11], and hip-fracture models leaned on scanner and patient variables [12]. In PGUPharyngitis, each city used a

different phone, and bacterial prevalence may differ between cities. A model could therefore score well by recognising the phone.

We set out to build a symptom-aware image model and, in doing so, to test for device and site shortcuts. Our contributions are:

1. The first multimodal benchmark on PGUPharyngitis, comparing symptom-only, image-only, clinical-score and five fusion approaches, including a gated cross-modal fusion network, under one repeated, nested cross-validation protocol with public folds.
2. A shortcut analysis showing that the phone alone, or 18 global colour statistics, match the published image baseline; that image features identify the phone almost perfectly; and that image performance largely disappears within a phone and across phones.
3. Ablation, label-dependence and tie-rule sensitivity analyses showing which conclusions hold under different analysis choices and which do not.
4. Open code, fold assignments and out-of-fold predictions to make future work on this dataset comparable.

2. Related work

Clinical scores. The Centor score assigns one point each for fever, tonsillar exudate, tender anterior cervical lymphadenopathy and absence of cough [3]; McIsaac added an age adjustment [4]. Both are used to decide on testing or empirical antibiotics, and both need findings from a physical examination. PGUPharyngitis records fever and cough but not exudate or lymph nodes, so only a modified score can be computed.

Throat-image AI. Yoo et al. trained convolutional networks on smartphone throat photographs to separate severe pharyngitis from normal throats [5], and Askarian et al. used colour and texture features from smartphone images to detect streptococcal pharyngitis [6]. These studies were small and from single sites. Shojaei et al. released PGUPharyngitis with image-only baselines evaluated by three-fold cross-validation, reporting AUCs from 0.554 (MobileNetV3) to 0.645 (DenseNet121) on the full dataset [7]. To our knowledge, no study has fused these images with symptoms or examined what the models learn.

66 *Multimodal fusion.* Medical image–tabular fusion is usually grouped into
 67 early fusion (concatenating features), late fusion (combining predictions) and
 68 joint or intermediate fusion (learning a shared representation) [8]. Gated
 69 units learn how much to trust each modality per sample [13], and feature-
 70 wise linear modulation (FiLM) lets one modality scale and shift the features
 71 of another [14]. We implement all of these families on equal terms.

72 *Shortcut learning and site confounding.* Shortcut learning occurs when a
 73 model relies on cues that are predictive in the training distribution but not
 74 causally related to the target [9]. In medical imaging, such cues include
 75 the hospital, scanner, positioning markers and patient metadata [10, 11, 12].
 76 Hidden stratification, where performance is driven by an easy subgroup, can
 77 make overall metrics misleading [15]. Methodological reviews recommend
 78 evaluation across sites, explicit confounder baselines and reporting of variance
 79 across splits [16]. We apply these recommendations to throat imaging.

80 3. Materials and methods

81 The study is reported following TRIPOD+AI [17]. All analyses are ex-
 82 ploratory: the analysis plan was written before modelling, but hypotheses
 83 were not formally frozen before the first results were seen. Figure 1 sum-
 84 marises the study design.

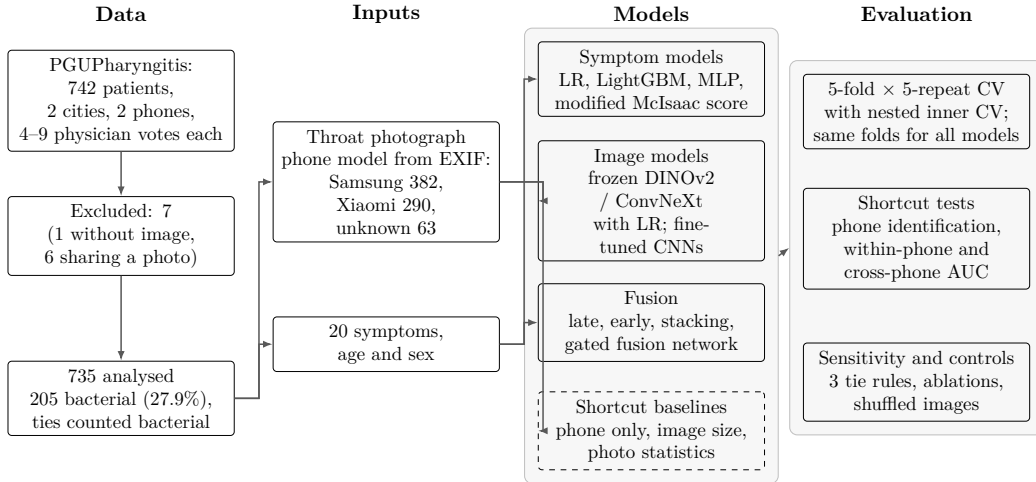


Figure 1: Study design: 742 patients, 735 analysed; inputs (photograph, 20 symptoms, age, sex), model families, 5×5 nested cross-validation and the shortcut analyses.

3.1. Dataset

We used PGUPharyngitis (Figshare, <https://doi.org/10.6084/m9.figshare.28163513>, CC BY 4.0) [7]. It contains one throat photograph per patient, taken with a Samsung Galaxy S21 Ultra or a Xiaomi phone in two Iranian cities between October 2023 and May 2024, and a table with age, sex, 20 binary symptoms and four to nine physician diagnoses (bacterial or non-bacterial). The original study received ethics approval (IR.BPUMS.REC.1403.282) with written informed consent; we used the anonymised public release only.

The table lists 742 patients. Patient 666 had no image. Three pairs of patients (19/23, 218/256, 301/305) shared an identical photograph (perceptual-hash distance 0) but had different ages, sexes or symptoms; because the image could not be assigned to either record, we excluded all six. The analysis set has 735 patients. The table has no city or phone column, so we read the phone model from each photograph’s EXIF metadata: Samsung (model code SM-G998B) for 382 patients, Xiaomi (2201117SG) for 290, and unknown for 63. Because each city used one phone [7], we treat the phone as a proxy for acquisition site and write “phone/site” throughout. Treatments were not recorded. The sample size was fixed by the public dataset and no sample size calculation was done; with 205 bacterial cases, AUC confidence intervals are about ± 0.04 wide, and we report them throughout.

3.2. Labels

The 735 patients received 3,662 diagnoses (median 5 per patient; one patient had 3 recorded). The reference label is the majority vote. One hundred patients (13.6%) had a tied vote; in the primary analysis ties count as bacterial, giving 205 bacterial cases (27.9%). Two sensitivity analyses count ties as non-bacterial (105 bacterial, 14.3%) or exclude them (635 patients, 16.5% bacterial). We also keep the soft label, the share of physicians voting bacterial, and the agreement, $|\text{soft} - 0.5| \times 2$; 230 patients (31%) had unanimous votes. The labels reflect expert consensus from photographs and symptoms, not microbiological testing. The data paper does not report the physicians’ specialties or whether they were blinded to each other’s diagnoses [7].

3.3. Preprocessing

Images were rotated according to EXIF orientation, converted to RGB and resized so the short side was 256 pixels. Training augmentation used a random resized crop to 224×224 (scale 0.8–1.0), horizontal flip, rotation up

121 to 15° and brightness and contrast jitter of 10%. Hue and saturation were
 122 not altered, because throat redness is diagnostic. At test time we used a 224-
 123 pixel centre crop and averaged predictions over the original and a horizontally
 124 flipped image. Images were normalised with ImageNet statistics. Tabular
 125 inputs were the 20 symptoms, sex (0/1) and age; age was standardised and
 126 any missing value imputed using statistics from the training data only (no
 127 values were missing).

128 3.4. Evaluation protocol

129 All models used the same stratified 5-fold cross-validation repeated five
 130 times with different seeds (25 test folds of 147 patients), stratified on label
 131 $\times$ phone. Fold assignments are released with the code. Within each training
 132 part, a 3-fold inner cross-validation selected hyperparameters (by AUC), the
 133 number of training epochs for neural fusion models, and the decision thresh-
 134 old, chosen to maximise balanced accuracy on inner out-of-fold predictions.
 135 Scalers, imputers, PCA, thresholds and stacking meta-models were fitted on
 136 training data only. Because fine-tuning is costly, fine-tuned CNNs used one
 137 repeat (5 folds) and a stratified 15% validation split of each training part for
 138 early stopping and threshold choice.

139 3.5. Models

140 *Symptom models.* Logistic regression (L2, class-balanced, $C \in \{0.01, 0.1, 1, 10\}$),
 141 LightGBM [18] (4 or 8 leaves, 100 or 300 trees, learning rate 0.05) and a
 142 two-layer perceptron (64–64 units, L2 penalty $\in \{0.001, 0.1, 1\}$) used the 20
 143 symptoms, sex and age.

144 *Clinical score.* A modified McIsaac score summed fever (“Pyrexia”), absence
 145 of cough and an age term (+1 for 3–14 years, -1 for ≥ 45 years) [4]; exudate
 146 and lymph-node items are not recorded.

147 *Reference baselines.* To measure shortcuts we trained three models that do
 148 not use throat anatomy: (i) *phone only*, predicting the training-fold bacterial
 149 rate of the photograph’s phone; (ii) *image size*, logistic regression on the log
 150 width, height, aspect ratio and area of the original photograph; and (iii) *photo*
 151 *statistics*, logistic regression on 18 global statistics of each resized photograph
 152 (RGB and HSV means and standard deviations, shares of dark, bright and
 153 strongly red pixels, and edge density).

154 *Frozen image models.* We extracted embeddings from DINOv2 ViT-S/14
 155 [19] (384 dimensions, 224-pixel input) and ConvNeXt-Tiny pretrained on
 156 ImageNet-22k and fine-tuned on ImageNet-1k [20] (768 dimensions), once per
 157 image for the test view, the flipped view and ten random training augmen-
 158 tations. Standardised embeddings were classified by class-balanced logistic
 159 regression ($C \in \{0.001, 0.01, 0.1, 1\}$).

160 *Fine-tuned image models.* To reproduce the published baselines we fine-tuned
 161 DenseNet121 [21], MobileNetV3-Large [22] and ConvNeXt-Tiny end to end
 162 with AdamW (learning rate 10^{-4} , weight decay 10^{-4} , cosine schedule, batch
 163 16), class-weighted cross-entropy, up to 30 epochs and early stopping on
 164 validation AUC (patience 7).

165 *Standard fusion.* *Late fusion* averaged the symptom and image logistic-regression
 166 probabilities. *Early fusion* fitted one logistic regression on the tabular inputs
 167 plus a PCA projection of the image embedding (16 or 64 components). *Stack-*
 168 *ing* fitted a logistic regression on the logits of inner out-of-fold symptom and
 169 image predictions.

170 *Gated cross-modal fusion.* The image embedding passes through layer nor-
 171 malisation, dropout and a linear layer to a 128-dimensional vector h_{img} ; the
 172 22 tabular inputs pass through a 22–64–128 perceptron to h_{sym} . A gate
 173 computes a scalar weight per patient,

$$g = \sigma(W[h_{\text{img}}; h_{\text{sym}}] + b), \quad z = g h_{\text{img}} + (1 - g) h_{\text{sym}}, \quad (1)$$

174 and a dropout–linear head maps z to a logit. Auxiliary image-only and
 175 symptom-only heads add their losses with weight λ , and modality dropout
 176 (probability 0.2 per branch, never both) randomly removes one branch during
 177 training. Training used the soft label as target in class-weighted binary cross-
 178 entropy, AdamW with weight decay 10^{-4} , batch 32 and a cosine schedule over
 179 60 epochs. Each epoch drew one of the ten cached augmentation embeddings
 180 per patient. The inner cross-validation selected the learning rate $\{10^{-3}, 3 \times$
 181 $10^{-4}\}$, dropout $\{0.3, 0.5\}$, $\lambda \in \{0, 0.3\}$ and the number of epochs (maximum
 182 of the smoothed mean inner AUC curve); the model was then retrained on
 183 the full training part. Ablations replaced the gate by concatenation or by
 184 FiLM [14], used hard labels, removed the auxiliary heads, removed modality
 185 dropout, or removed fever and cough from the inputs.

3.6. Shortcut and label-dependence analyses

1. *Phone identification*: logistic regression predicting Xiaomi versus Samsung (672 patients with known phone) from image embeddings, image size or photo statistics, in 5-fold cross-validation.
2. *Within-phone evaluation*: for each model, AUC computed separately within the Samsung, Xiaomi and unknown-phone strata of the out-of-fold predictions and averaged weighted by stratum size. A phone-only model scores 0.5 by construction.
3. *Within-phone training*: models trained and tested inside one phone (5-fold $\times$ 5 repeats).
4. *Cross-phone transfer*: models trained on all patients of one phone and tested on the other, with patient-level bootstrap confidence intervals (1,000 resamples).
5. *Large images only*: within-phone AUC restricted to photographs whose original short side was at least 500 pixels.
6. *Shuffled-image control*: gated fusion trained after permuting image embeddings across patients; with no leakage it should lose the image signal.
7. *Label dependence*: AUC on unanimous versus split votes; Spearman correlation with the vote share; and a residual test in which DINOv2 embeddings predict the out-of-fold errors of the symptom model (ridge regression, 5-fold cross-validated correlation, 500 permutations).

3.7. Statistics

For each model and repeat, the five test folds were pooled and AUC, PR-AUC, balanced accuracy, macro F1, sensitivity, specificity, Brier score and calibration slope were computed; we report the mean $\pm$ SD across repeats. AUC 95% confidence intervals come from 1,000 patient-level bootstrap resamples within each repeat, averaged across repeats. Model comparisons used the Nadeau–Bengio corrected resampled t -test on the 25 fold-level AUC differences [23], which accounts for overlapping training sets, and the DeLong test [24, 25] within each repeat. Three hypotheses compared gated fusion with the best symptom-only model (H1), the best image-only model (H2) and the best standard fusion model (H3), one-sided, with Holm correction [26]; choosing the best comparator by its own performance makes these tests conservative. Decision curves [27] show net benefit for threshold probabilities 0.05–0.60. To check fairness, we report discrimination by sex and age band; no fairness corrections were applied.

223 3.8. Explainability and implementation

224 We computed SHAP values [28] for a LightGBM symptom model, sum-
 225 marised gate values by vote agreement, and produced Grad-CAM maps [29]
 226 from a ConvNeXt-Tiny fine-tuned on all patients for the median best epoch
 227 count (seven), used only for visualisation. Code ran on an Apple M5 Mac-
 228 Book Air (16 GB) with Python 3.11, PyTorch 2.14 (Metal backend), timm
 229 1.0, scikit-learn 1.9 and LightGBM 4.7.

230 4. Results

231 4.1. Cohort

232 The 735 patients had a mean age of 28.8 years (SD 17.1, range 2–83)
 233 and 373 (50.7%) were male (Table 1). Bacterial prevalence differed threefold
 234 between phones: 14.9% for Samsung photographs and 43.8% for Xiaomi
 235 photographs.

Table 1: Cohort characteristics (primary analysis, ties counted as bacterial).

Characteristic	Value
Patients analysed	735 (of 742)
Age, mean (SD), range	28.8 (17.1), 2–83 years
Male	373 (50.7%)
Physician diagnoses, total (median per patient)	3,662 (5)
Unanimous votes	230 (31.3%)
Tied votes	100 (13.6%)
Bacterial, majority vote	205 (27.9%)
Samsung photographs: patients (bacterial)	382 (57; 14.9%)
Xiaomi photographs: patients (bacterial)	290 (127; 43.8%)
Phone unknown: patients (bacterial)	63 (21; 33.3%)

236 4.2. Benchmark

237 Table 2 and Fig. 2 summarise the benchmark. The best model was stacked
 238 fusion of DINOv2 image and symptom predictions (AUC 0.703, 95% CI 0.66–
 239 0.74), followed by the frozen DINOv2 image model alone (0.700). Late, early
 240 and gated fusion reached 0.683–0.696. Fine-tuning did not beat frozen fea-
 241 tures: ConvNeXt-Tiny reached 0.688, DenseNet121 0.672 and MobileNetV3

0.578, close to the published 0.645 and 0.554 [7]. Symptoms alone were weak (best 0.596), and the modified McIsaac score ranked patients worse than chance (0.437).

Three models that use no throat anatomy did about as well as the published deep networks. The phone-only model reached 0.673, above the published image baseline and above all symptom models. Eighteen global photo statistics reached 0.646, and the original image size 0.562.

Table 2: Discrimination under 5×5 cross-validation (fine-tuned CNNs: 1×5). “Within phone” is the size-weighted mean AUC inside each phone stratum. Balanced accuracy and macro F1 use thresholds chosen by inner cross-validation.

Model	AUC [95% CI]	Within phone	Bal. acc.	F1	Brier
Stacked fusion, DINOv2 + sympt.	0.703 [0.66, 0.74]	0.586	0.659	0.637	0.218
Image, frozen DINOv2	0.700 [0.66, 0.74]	0.586	0.656	0.630	0.215
Gated fusion, DINOv2	0.691 [0.65, 0.73]	0.584	0.645	0.613	0.188
Image, fine-tuned ConvNeXt-Tiny	0.688 [0.64, 0.73]	0.605	0.652	0.615	0.211
Gated fusion, ConvNeXt	0.684 [0.64, 0.73]	0.582	0.624	0.606	0.190
Image, frozen ConvNeXt-Tiny	0.681 [0.64, 0.72]	0.561	0.641	0.605	0.218
Phone only	0.673 [0.63, 0.71]	0.488	0.668	0.622	0.182
Image, fine-tuned DenseNet121	0.672 [0.63, 0.71]	0.556	0.622	0.562	0.207
Photo statistics (18 features)	0.646 [0.60, 0.69]	0.536	0.598	0.559	0.233
Symptoms, perceptron	0.596 [0.55, 0.64]	–	0.578	0.563	0.236
Symptoms, LightGBM	0.591 [0.54, 0.64]	0.550	0.561	0.537	0.240
Image, fine-tuned MobileNetV3	0.578 [0.53, 0.62]	0.519	0.550	0.542	0.324
Image size only	0.562 [0.51, 0.61]	0.493	0.536	0.504	0.247
Gated fusion, shuffled images	0.529 [0.48, 0.57]	0.507	0.524	0.500	0.211
Modified McIsaac score	0.437 [0.39, 0.48]	–	0.493	0.245	0.241

4.3. Device and site shortcuts

Every analysis pointed to the phone as the main source of image performance (Table 3, Fig. 3). Image embeddings identified the phone almost perfectly (AUC 0.993 for DINOv2 and 0.997 for ConvNeXt), and so did the 18 photo statistics (0.958). When AUC was computed within each phone,

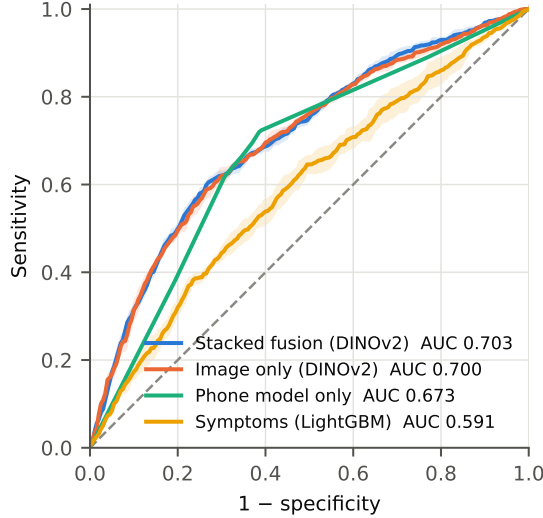


Figure 2: Mean ROC curves (± 1 SD over five repeats) for stacked fusion (DINOv2), the frozen DINOv2 image model, the phone-only model and the LightGBM symptom model. Dashed line: chance.

254 image models fell from about 0.70 to 0.56–0.61, while the phone-only model
 255 fell to 0.49, as expected. Models trained and tested inside one phone did
 256 worse still: image AUC was 0.50 on Samsung photographs and 0.55–0.58 on
 257 Xiaomi photographs. Models did not transfer between phones: trained on
 258 one phone and tested on the other, image AUC was 0.50–0.53 and every 95%
 259 confidence interval included 0.5. Restricting to photographs with an original
 260 short side of at least 500 pixels did not change within-phone AUC (DINOv2:
 261 0.568 Samsung, 0.611 Xiaomi), so very small crops do not explain the effect.

262 The frozen DINOv2 image model beat the phone-only model by 0.025
 263 AUC (corrected t -test $p = 0.035$, uncorrected), whereas ConvNeXt did not
 264 (+0.006, $p = 0.35$). Within the Xiaomi subgroup the photographs therefore
 265 carry a modest signal (AUC about 0.58–0.61); within the Samsung subgroup
 266 they carry essentially none.

267 4.4. Does fusion help?

268 Gated fusion (DINOv2) beat the best symptom model by 0.095 AUC
 269 (Holm-adjusted $p = 0.012$; significant in 5 of 5 DeLong repeats) but did not
 270 beat the best image-only model (-0.020 , $p = 1.0$) or the best standard fusion
 271 (-0.008 , $p = 1.0$; Table 4). Stacking added 0.003 AUC over the image alone

Table 3: Shortcut analyses (AUC). In the cross-phone rows, the column names the phone used for testing; the model was trained on the other phone.

Analysis	Samsung	Xiaomi	Overall
Identify phone: DINOv2 / ConvNeXt embeddings	–	–	0.993 / 0.997
Identify phone: photo statistics / image size	–	–	0.958 / 0.754
Within-phone training: DINOv2 image	0.502	0.575	–
Within-phone training: ConvNeXt image	0.499	0.548	–
Within-phone training: DINOv2 + symptoms	0.546	0.561	–
Within-phone training: symptoms	0.627	0.517	–
Within-phone training: photo statistics	0.507	0.538	–
Cross-phone: DINOv2 image [95% CI]	0.502 [0.42, 0.59]	0.519 [0.45, 0.59]	–
Cross-phone: ConvNeXt image [95% CI]	0.507 [0.42, 0.59]	0.530 [0.47, 0.60]	–
Cross-phone: symptoms [95% CI]	0.424 [0.34, 0.52]	0.482 [0.41, 0.55]	–

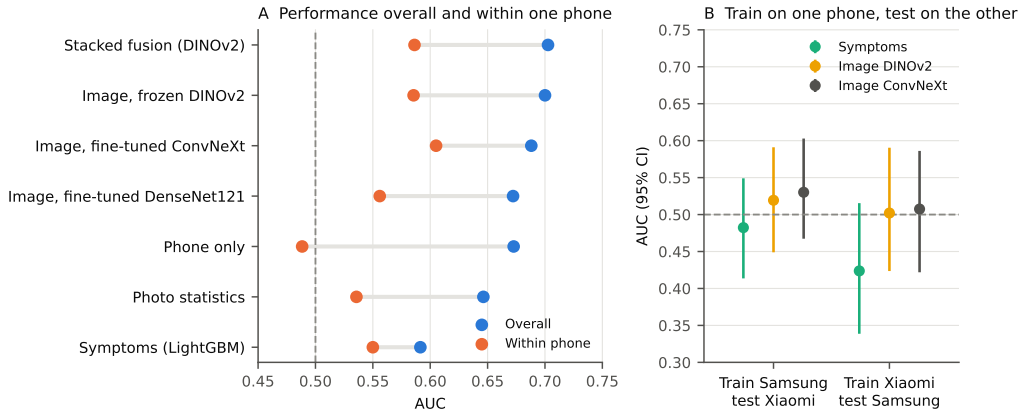


Figure 3: Device and site shortcut. (A) AUC over all patients and the size-weighted AUC within each phone; the phone-only model drops to chance within a phone. (B) Models trained on all photographs from one phone and tested on the other, with 95% bootstrap confidence intervals; all intervals include 0.5.

($p = 0.29$). Within a phone, adding symptoms to the image changed AUC by -0.01 to $+0.04$. Symptoms therefore add little once the photograph is known.

Table 4: Planned comparisons for gated fusion (DINOv2) under three tie rules. Δ AUC is the mean fold-level difference; p values are one-sided, Nadeau–Bengio corrected and Holm-adjusted.

Comparison	Ties bacterial	Ties non-bacterial	Ties dropped
H1: vs best symptom-only	$+0.095$ ($p = 0.012$)	$+0.044$ ($p = 0.38$)	$+0.051$ ($p = 0.22$)
H2: vs best image-only	-0.020 ($p = 1.0$)	-0.009 ($p = 1.0$)	-0.002 ($p = 1.0$)
H3: vs best standard fusion	-0.008 ($p = 1.0$)	-0.018 ($p = 1.0$)	-0.019 ($p = 1.0$)
Image (DINOv2): AUC / within phone	0.700 / 0.586	0.671 / 0.609	0.696 / 0.605
Phone only: AUC	0.673	0.618	0.650
Best symptom-only: AUC	0.596	0.616	0.639

4.5. Ablations

No component of the fusion network made a measurable difference (Fig. 4). With ConvNeXt features, the full gated model reached 0.684; replacing the gate by concatenation gave 0.686 ($p = 0.60$), FiLM 0.681, removing the auxiliary heads 0.682 and removing modality dropout 0.682. Training on hard instead of soft labels lowered AUC to 0.672 (difference 0.012, $p = 0.19$). Removing fever and cough did not hurt (0.690). The shuffled-image control fell to 0.529, below symptoms alone. The pipeline therefore does not leak labels, and the fusion network relied almost entirely on the image branch.

4.6. Label dependence and sensitivity

Models discriminated better on unanimous than on split votes (DINOv2 image: 0.719 vs 0.678; symptoms: 0.623 vs 0.580). Image predictions correlated more with the physicians’ vote share than symptom predictions did (Spearman 0.32 vs 0.13), though the phone-only model also reached 0.28. Image embeddings predicted the out-of-fold errors of the symptom model (cross-validated $r = 0.29$; permutation $p = 0.002$), but this test cannot separate throat appearance from phone identity. Under both alternative tie rules

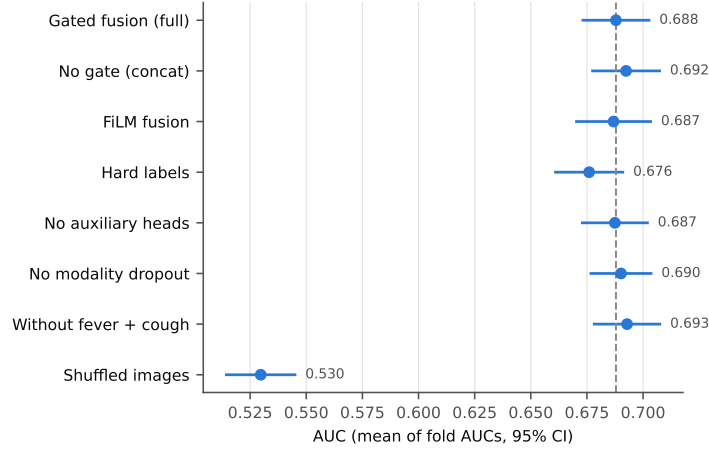


Figure 4: Ablation of the gated fusion network (ConvNeXt features): mean fold AUC with 95% confidence interval. The shuffled-image control falls to chance.

the conclusions held: image models stayed near 0.67–0.70 overall and about 0.61 within phone, fusion never beat the image alone, and the advantage of fusion over symptoms was no longer significant (Table 4).

4.7. Calibration and clinical utility

Among the frozen-feature image and fusion models, Brier scores ranged from 0.188 (gated fusion, DINOv2) to 0.226 (stacking, ConvNeXt); the phone-only model scored 0.182. Class-balanced training made image and fusion models overestimate risk, and decision-curve analysis showed no net benefit over treating everyone at thresholds up to 0.20; at 0.30 the stacked model gave a net benefit of 0.024 versus -0.030 for treat-all, but the phone-only model gave 0.082 (Fig. 5). The models are therefore not ready for treatment decisions.

4.8. Explainability and subgroups

In the symptom model, cough, age, vomiting, myalgia and sore throat had the largest mean absolute SHAP values (Fig. A.8). The gate weighted the image at $g \approx 0.48$ on average (DINOv2) and slightly less when physicians disagreed (median 0.47 vs 0.50, Mann–Whitney $p < 0.001$; Fig. A.9). Grad-CAM maps (Fig. 6) of the most confident bacterial cases highlighted white spots in close-up photographs, whereas the most confident non-bacterial cases were wide views of the oral cavity with the uvula and tongue visible. Framing

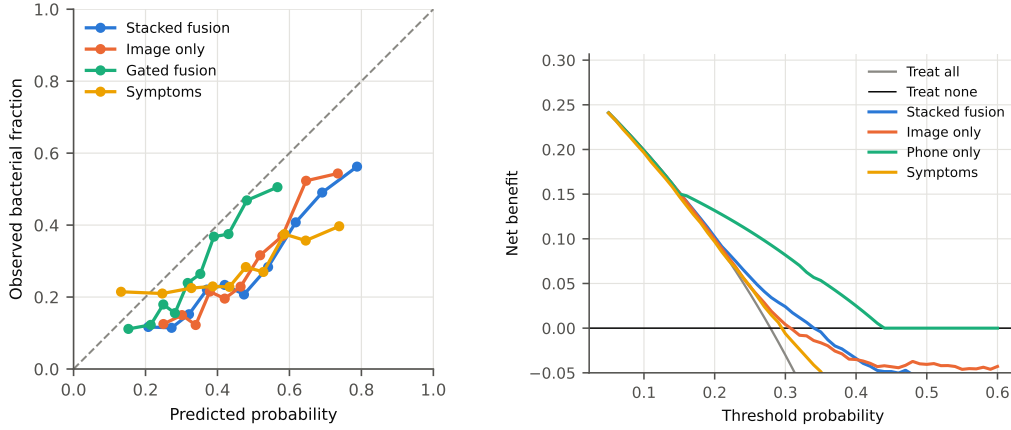


Figure 5: (A) Calibration curves and (B) decision curves for the main models. Treat-all and treat-none are shown for reference.

and zoom, which may differ between sites, may therefore act as an additional
 cue. The image model discriminated better in children aged 14 or younger
 (AUC 0.77) than in adults aged 45 or older (0.64), and similarly in women
 (0.68) and men (0.72) (Fig. A.7).

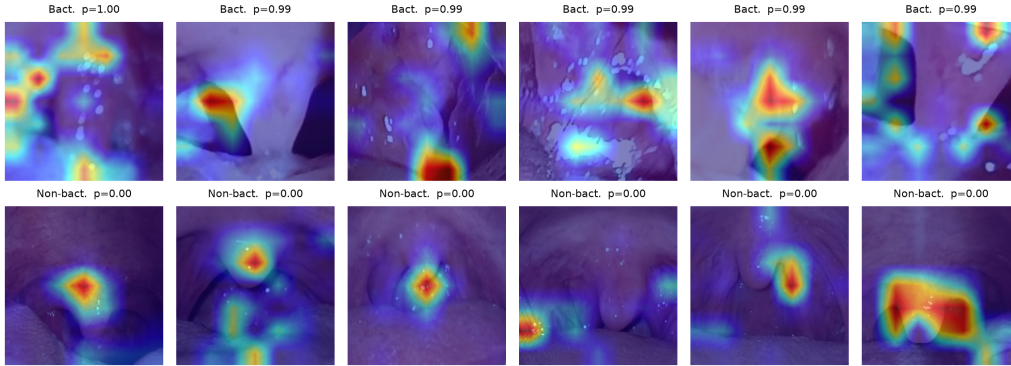


Figure 6: Grad-CAM maps of a ConvNeXt-Tiny fine-tuned on all patients for the six most confident bacterial (top) and non-bacterial (bottom) cases. Red marks high attention. Bacterial predictions are close-ups; non-bacterial predictions are wide views.

316 5. Discussion

317 We set out to improve smartphone pharyngitis classification by adding
318 symptoms to throat photographs. Instead, the benchmark showed that most
319 of the apparent image performance on PGUPharyngitis comes from recog-
320 nising the phone, and with it the city, that produced the photograph. The
321 phone alone predicted the expert label as well as the published deep-learning
322 baseline, and 18 colour statistics nearly matched it. Image embeddings iden-
323 tified the phone almost perfectly; within one phone, image models lost most
324 of their discrimination; and no model transferred between phones.

325 *Why the shortcut exists.* Bacterial prevalence was 44% among Xiaomi pho-
326 tographs and 15% among Samsung photographs. Whether this reflects true
327 epidemiological differences between the two cities, different referral patterns
328 or different labelling physicians cannot be resolved from the released data.
329 Either way, any feature that identifies the phone (colour rendering, sharp-
330 ness, lighting, framing) becomes predictive of the label. Our Grad-CAM
331 maps suggest that framing differs too: confident bacterial predictions were
332 close-ups, confident non-bacterial predictions wide views. This is the same
333 failure mode reported for chest radiographs, where models recognised hospi-
334 tals and scanners [10, 11, 12], and it is invisible in pooled cross-validation.

335 *What remains when the shortcut is removed.* Within the Xiaomi subgroup,
336 image models kept a modest signal (AUC 0.58–0.61); within the Samsung
337 subgroup, which had only 57 bacterial cases, they kept essentially none. The
338 images do contain some information beyond symptoms, as the residual test
339 showed, but a within-site AUC around 0.6 is far from clinically useful. The
340 image model also worked best in children, in whom streptococcal pharyngitis
341 is most common, which may deserve a dedicated study.

342 *Fusion.* Fusion beat symptoms alone only under one of three equally defen-
343 sible tie rules, and it never beat the image alone. The gate, auxiliary heads,
344 modality dropout and FiLM conditioning made no difference, and the gated
345 model learned to weight both branches almost equally. With symptoms this
346 weak (AUC about 0.6) and an image signal dominated by site, there is little
347 for a fusion architecture to exploit. For future work on this dataset, the
348 lesson is that a better fusion architecture will not fix confounded data.

349 *Symptoms and clinical scores.* The modified McIsaac score ranked patients
 350 worse than chance. Two explanations are likely: the dataset lacks exudate
 351 and lymph-node items, and the labels come from physicians who were not
 352 using a clinical score. Cough and age were the most informative symptoms,
 353 consistent with clinical scores, but together the symptoms explained little of
 354 the physicians’ decision.

355 *Implications.* Smartphone throat-imaging studies should (i) report a device-
 356 or site-only baseline, (ii) evaluate within and across acquisition sites, (iii)
 357 balance or randomise devices across sites at data collection, and (iv) stan-
 358 dardise framing. Datasets should record site and device per image. A de-
 359 ployable model would also need on-device checks of photograph quality and
 360 framing, since the models here receive a single photograph and 22 tabular
 361 inputs and give no warning when a photograph is unusable; users would need
 362 only to take the photograph and record the symptoms. For PGUPharyngi-
 363 tis, we recommend that future work report within-phone and cross-phone
 364 AUC alongside pooled results; our folds and out-of-fold predictions make
 365 such comparisons straightforward.

366 5.1. Limitations

- 367 1. Labels are physician consensus from photographs and symptoms, not
 368 throat culture or rapid antigen tests, so all results concern agreement
 369 with experts rather than infection.
- 370 2. City is not recorded; we used the EXIF phone model as a proxy for site
 371 and cannot separate phone from city effects. Sixty-three photographs
 372 had no EXIF phone.
- 373 3. One hundred patients had tied votes; conclusions about fusion versus
 374 symptoms depend on the tie rule.
- 375 4. The dataset is small (205 bacterial cases) and from one country, and
 376 there is no external test set.
- 377 5. Fine-tuned CNNs were evaluated with one repeat instead of five, so
 378 their estimates are less precise.
- 379 6. The analyses were planned but not formally pre-registered and should
 380 be regarded as exploratory.
- 381 7. Grad-CAM maps were not rated by clinicians, and framing differences
 382 were observed, not measured.

383 6. Conclusion

384 On the largest public smartphone pharyngitis dataset, image models ap-
385 pear to reach an AUC near 0.70, but most of this performance comes from
386 recognising the phone and site that took the photograph. Within one site
387 the photographs carry at most a modest signal, nothing transfers across sites,
388 and adding symptoms through any fusion design does not help. Smartphone
389 pharyngitis models should be judged by their within-site and cross-site per-
390 formance, against a device-only baseline, before any clinical claim is made.

391 CRediT authorship contribution statement

392 **Sandesh Bhatta:** Conceptualization, Methodology, Software, Valid-
393 ation, Formal analysis, Investigation, Data curation, Visualization, Writing –
394 original draft, Writing – review & editing.

395 Declaration of competing interest

396 The author declares no known competing financial interests or personal
397 relationships that could have appeared to influence the work reported in this
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402 Ethics statement

403 This study used only the anonymised, publicly released PGUPharyngi-
404 tis dataset. The original data collection was approved by a research ethics
405 committee (IR.BPUMS.REC.1403.282) with written informed consent from
406 all participants [7]. No additional ethics approval was required for this sec-
407 ondary analysis of public data.

408 Protocol, registration and patient involvement

409 No formal study protocol was published and the study was not registered;
410 the analysis decisions are recorded in the code repository (`paper/analysis_plan.md`).
411 Patients and the public were not involved in the design, conduct or reporting
412 of this study.

413 **Data availability**

414 PGUPharyngitis is publicly available under CC BY 4.0 at <https://doi.org/10.6084/m9.figshare.28163513>. Fold assignments and out-of-fold
415 predictions are released with the code.
416

417 **Code availability**

418 All code, configuration files, fold assignments and out-of-fold predictions
419 are available at <https://github.com/sandesh-bhatta-9/pharyngitis-fusion>
420 under the MIT licence. Two scripts reproduce every table and figure.

421 **Declaration of generative AI and AI-assisted technologies in the 422 writing process**

423 During the preparation of this work the author used Claude (Anthropic)
424 to assist with drafting and editing the manuscript. After using this tool, the
425 author reviewed and edited the content as needed and takes full responsibility
426 for the content of the publication.

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430 **Appendix A. Supplementary figures**

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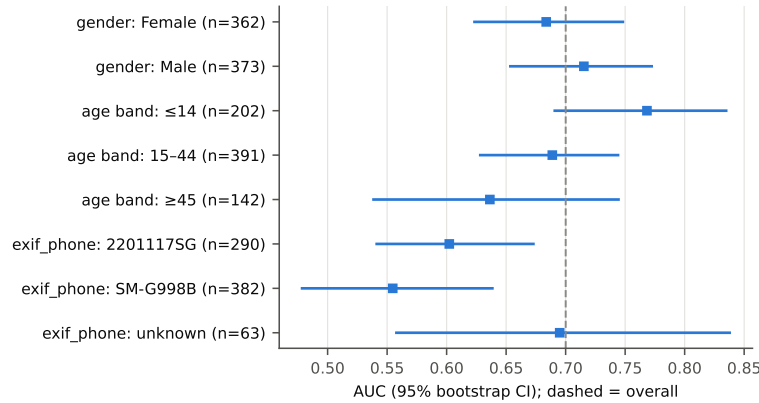


Figure A.7: AUC of the frozen DINOv2 image model by sex, age band and phone, with 95% bootstrap confidence intervals. Dashed line: overall AUC.

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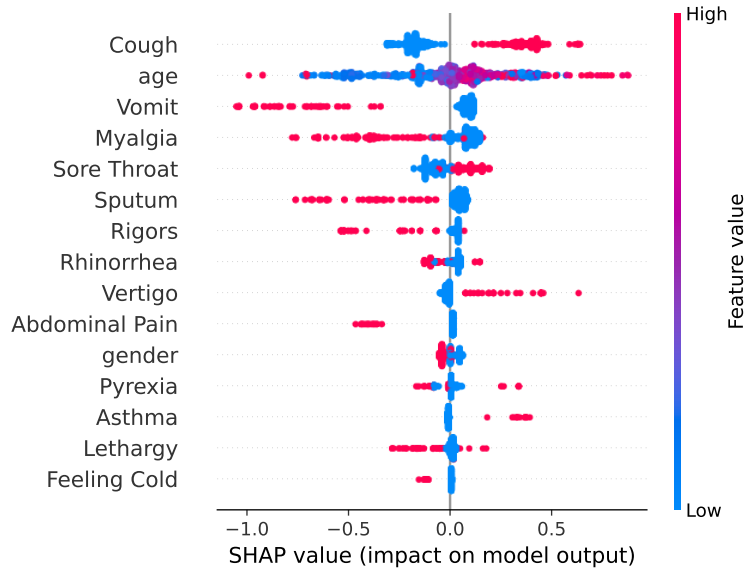


Figure A.8: SHAP summary of the LightGBM symptom model fitted on all patients.

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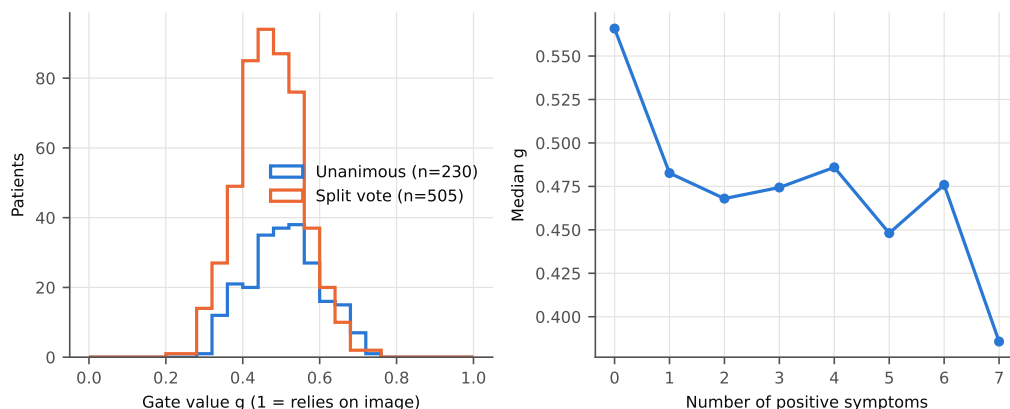


Figure A.9: Gate values of the gated fusion model (DINOv2): distribution for unanimous and split votes (left) and median gate value by number of positive symptoms (right). $g = 1$ means the model relies only on the image.

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